

FLAME-US: A Landmark-Guided Dataset for Quantitative Fetal Motion Analysis in Freehand Ultrasound

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Abstract. Fetal movement, particularly prominent during second-trimester ultrasound (US) examinations poses a major challenge for motion-aware applications such as 3D reconstruction, standard-plane acquisition, and robotic US navigation. Reliable quantification of fetal motion is therefore essential. However, automated fetal motion analysis remains fundamentally ill-posed because frame-to-frame image changes arise from two independent and entangled sources: probe motion and fetal motion. While probe motion can be measured using external tracking systems, fetal pose cannot be directly observed during routine scanning, resulting in a lack of datasets with explicit fetal-motion supervision. To address this gap, we present FLAME-US, a data-engineering framework that generates explicit fetal-motion supervision by simulating independent fetal and probe motions within patient-derived 3D fetal head volumes, producing trajectories with known fetal and probe Special Euclidean Group (SE(3)) poses. We further introduce Anatomical Landmark Residuals (ALRs), defined as per-frame landmark displacements attributable exclusively to fetal motion and decoupled from probe movement. Built from 25 patient-derived fetal head volumes, FLAME-US generates an open-source dataset of 10,000 motion trajectories annotated with fetal and probe poses and seven ALRs. Validation shows that ALRs strongly correlate with fetal motion magnitude (Pearson correlation coefficient $r = 0.74$, $p < 0.001$). Baseline experiments demonstrate that the generated annotations are learnable, achieving fetal rotation error of 0.438 radians, an 85.7% reduction in error from the lowest to highest motion-magnitude quintile. By providing explicit fetal-motion supervision where direct measurement is not possible, FLAME-US enables future learning-based approaches for fetal motion quantification and motion-aware ultrasound systems.

Keywords: Fetal Ultrasound · Key structure Segmentation · Fetal head Motion Quantification · Data Generation · Landmark Residuals

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1 Introduction

Fetal movement provides important insight into fetal wellbeing and neuromuscular development [18,9], but also poses a major challenge for ultrasound (US) applications such as 3D reconstruction [10], standard-plane acquisition [5], and autonomous robotic scanning [7]. Reliable quantification of fetal motion is therefore essential. However, automated fetal motion analysis from freehand 2D US remains fundamentally ill-posed because frame-to-frame image changes arise from two independent and entangled sources: fetal motion and probe motion. While probe motion can be measured during acquisition [13], fetal pose cannot be directly observed in routine scanning [17], resulting in a lack of explicit fetal-motion supervision despite the widespread clinical use of US.

Existing fetal US datasets have primarily focused on anatomical annotations such as segmentation masks, biometric landmarks, and standard-plane labels [4,1,6]. More recent fetal video datasets provide temporal image sequences but do not contain explicit fetal-motion labels or probe trajectory information [3,14]. Similarly, datasets developed for freehand 3D fetal US reconstruction often provide probe pose measurements but assume a stationary fetus, attributing all observed image variation to probe motion [11,12]. As a result, current data resources do not provide disentangled supervision of independently moving probe and fetal poses, limiting the development of learning-based methods for fetal motion quantification, motion modelling, and motion-aware navigation [19,20]. To address this gap, we present *FLAME-US*, a data-engineering framework that generates explicit fetal-motion supervision from patient-derived 3D fetal head volumes. FLAME-US simulates independent fetal and probe motions to produce large-scale trajectories with known fetal and probe Special Euclidean Group (SE(3)) poses. We further introduce *Anatomical Landmark Residuals* (ALRs), segmentation-derived landmark displacements that isolate fetal motion from probe movement, providing localized anatomical cues complementary to global pose supervision. By transforming patient-derived volumes into richly annotated motion trajectories with known fetal dynamics, FLAME-US creates a practical source of supervision, enabling future learning-based approaches for fetal motion analysis and motion-aware ultrasound systems. Our contributions are:

- **FLAME-US**. A framework for generating explicit fetal-motion supervision by simulating independent fetal and probe motions within patient-derived 3D fetal US volumes.
- **Anatomical Landmark Residuals (ALRs)**. A novel anatomy-aware representation of fetal motion based on segmentation-derived landmark displacements decoupled from probe movement.
- **An open-source fetal-motion benchmark**. 10,000 trajectories annotated with disentangled fetal and probe SE(3) poses and seven ALRs, validated for motion realism and supervision learnability.

2 Methodology

FLAME-US is a data engineering pipeline designed to disentangle fetal motion from probe motion in consecutive 2D US frames. Given a pair of consecutive frames (I_i, I_{i+1}) and tracked probe poses, the pipeline generates paired SE(3) supervision targets for probe-relative motion $\mathbf{T}^{(i)}_{\text{probe,rel}}$, fetus-relative motion $\mathbf{T}^{(i)}_{\text{fetal,rel}}$, and combined relative motion $\mathbf{T}^{(i)}_{\text{total,rel}}$, together with anatomy-aware landmark residuals that capture localized fetal head motion. The overall dataset curation pipeline is illustrated in Fig. 1.

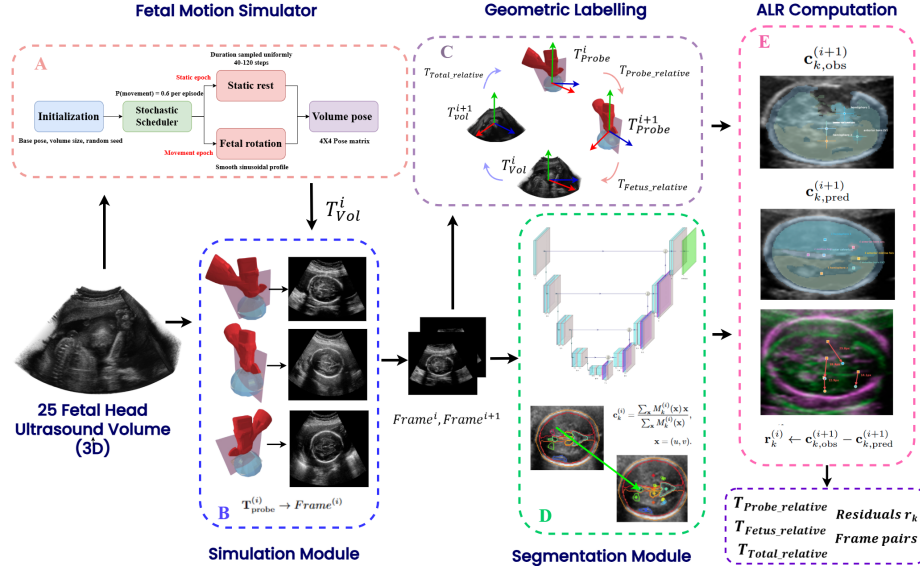


Fig. 1. Overview of the **FLAME-US** dataset curation pipeline. (A) A clinically grounded fetal motion simulator generates smooth fetal head motion. (B) Probe trajectories are simulated over real 3-D fetal head US volumes to generate 2-D frame sequences. (C) Generation of relative fetal, probe, and combined motion poses from consecutive frame pairs. (D) Anatomy-aware segmentation extracts fetal head landmarks and the centroids are computed. (E) ALRs are computed as the discrepancy between landmark locations predicted using only probe motion and the observed landmark locations at frame $(i+1)$, thereby capturing localized fetal motion cues.

Data Source and Acquisition FLAME-US is built from patient-derived 3D fetal head ultrasound (US) volumes acquired at a tertiary clinical centre using clinical 3D probes. Data acquisition focused on singleton pregnancies between 19–21 gestational weeks, a clinically important window that provides optimal fetal head visibility and anatomical development for standard plane assessment. Acquisition was limited by factors including fetal position, maternal wellbeing, sonographer time constraints, and reduced US penetration in high-BMI patients, resulting in approximately 4–6 suitable cases per 20 daily examinations. In total, 40 fetal head US volumes were collected. From these, 25 volumes were selected

based on image quality and anatomical completeness, including minimal shadowing and motion artifacts and clear visualization of the three standard fetal head planes with clear visibility of all relevant anatomical structures. These selected volumes were subsequently used for dataset curation.

A. Fetal Motion Simulator: Design and Implementation As ground-truth fetal pose trajectories are unavailable in clinical US, fetal motion was simulated based on expert sonographer feedback. Clinicians consistently described second-trimester fetal head motion as gradual and episodic, occurring over short intervals rather than as abrupt or highly erratic movements. Motion was observed predominantly as changes in head orientation, with rotations about the sagittal and coronal axes being more common than axial rotation. These observations are consistent with International Society of Ultrasound in Obstetrics and Gynecology (ISUOG) guidelines, which acknowledge the need for fetal head repositioning to obtain standard neurosonographic views [15]. Motivated by this behaviour, fetal motion was modelled using smooth sinusoidal rotations, providing a simple approximation of the continuous, bounded reorientation patterns observed during routine scanning. Each trajectory was partitioned into alternating STATIC and MOVING segments. During STATIC segments the fetal volume remained fixed, corresponding to positive and negative rotations about the sagittal and coronal axes. Rotations about the axial axis were excluded based on sonographer feedback indicating their relatively infrequent occurrence. Within each MOVING segment, the instantaneous rotation angle followed a smooth sinusoidal profile,

$$\theta(t) = \theta_{\max} \sin(\pi p_t), \quad (1)$$

where $p_t \in [0, 1]$ denotes the normalized progress through the segment. The angle increases smoothly from zero to $\theta_{\max}$ at mid-segment before returning to zero, yielding a continuous reorientation without abrupt transitions. The peak angle $\theta_{\max}$ is sampled per segment from a fixed range, $\theta_{\max} \sim \mathcal{U}(3.2^\circ, 8.0^\circ)$, with 8° enforced as a hard upper limit. This bound was chosen conservatively so that rotation about the volume centroid keeps the imaged anatomy within the field of view: we visually inspected rendered slices at peak-angle frames across representative trajectories and confirmed no boundary clipping or warping within this range.

B. Trajectory Simulation from Volumetric Data Probe trajectories are generated by simulating the scanning process over a 3D US volume. At each timestep, a probe pose $\mathbf{T}_{\text{probe}}^{(i)} \in \mathbb{SE}(3)$ in world coordinates defines a plane that intersects the volume; a 2D US frame is extracted at this plane via volumetric interpolation. An expert controller generates goal-directed probe actions that navigate toward anatomical target planes. The volume reference frame itself moves according to a fetal motion schedule given by the fetal motion simulator, with world pose $\mathbf{T}_{\text{vol}}^{(i)} \in \mathbb{SE}(3)$, and the effective image plane is determined by composing both transformations. This yields a sequence of synchronized observations: ordered 2D frames paired with their corresponding world-frame probe

pose $\mathbf{T}_{\text{probe}}^{(i)}$ and volume pose $\mathbf{T}_{\text{vol}}^{(i)}$. By construction, each frame implicitly encodes both probe-driven motion and fetus-driven motion, enabling supervised learning of motion decomposition from the trajectory data.

C. Geometric Labeling With both pose sequences known exactly, FLAME-US derives three pairwise supervision targets from the probe and fetal-volume world poses at each frame as detailed in the algorithm 0. When the fetus is static, $\mathbf{T}_{\text{total,rel}}^{(i)}$ reduces to probe motion alone. When the probe is static, it captures only fetal motion. In realistic scans both effects contribute jointly, while $\mathbf{T}_{\text{fetal,rel}}^{(i)}$ supplies direct motion decomposition supervision.

Algorithm 1 Geometric Labeling

Require: World poses $\mathbf{T}_{\text{probe}}^{(i)} \in \mathbb{SE}(3)$, $\mathbf{T}_{\text{vol}}^{(i)} \in \mathbb{SE}(3)$ for frames $i, i+1$

Ensure: Three pairwise supervision targets: $\mathbf{T}_{\text{probe,rel}}^{(i)}$, $\mathbf{T}_{\text{total,rel}}^{(i)}$, $\mathbf{T}_{\text{fetal,rel}}^{(i)}$

$$\begin{aligned}
 \mathbf{T}_{\text{probe,rel}}^{(i)} &\leftarrow (\mathbf{T}_{\text{probe}}^{(i)})^{-1} \mathbf{T}_{\text{probe}}^{(i+1)} &> \text{Tracked probe-relative motion} \\
 \mathbf{T}_{\text{in-vol}}^{(i)} &\leftarrow (\mathbf{T}_{\text{vol}}^{(i)})^{-1} \mathbf{T}_{\text{probe}}^{(i)} &> \text{Probe pose in fetal-volume frame} \\
 \mathbf{T}_{\text{total,rel}}^{(i)} &\leftarrow (\mathbf{T}_{\text{in-vol}}^{(i)})^{-1} \mathbf{T}_{\text{in-vol}}^{(i+1)} &> \text{Total reconstruction label} \\
 \mathbf{T}_{\text{fetal,rel}}^{(i)} &\leftarrow (\mathbf{T}_{\text{vol}}^{(i)})^{-1} \mathbf{T}_{\text{vol}}^{(i+1)} &> \text{Explicit fetal-motion target (sim only)}
 \end{aligned}$$

D. Segmentation Module and Landmark Detection To provide local evidence of fetal motion, FLAME-US extracts anatomical landmarks using a 2-D U-Net [16] fine-tuned on manually annotated clinical US frames. From 24 annotated landmarks (22 inner, 2 outer), we select 7 clinically stable structures for residual motion estimation, consistent with landmarks routinely used in fetal biometry and growth assessment [2], spanning two acoustic regimes: low-contrast inner structures (S1–S6) and the high-contrast inner calvarium (S7). To account for this heterogeneity, we employ a dual-head U-Net with a shared encoder: a inner structure head for S1–S6 trained using Dice and binary cross-entropy losses, and a dedicated calvarium head for S7 trained with a weighted boundary loss [8]. For each structure k , the predicted mask $\mathbf{M}_k^{(i)}$ yields a centroid $\mathbf{c}_k^{(i)}$, used as a local motion signal:

$$\mathbf{c}_k^{(i)} = \begin{bmatrix} \bar{u}_k^{(i)} \\ \bar{v}_k^{(i)} \end{bmatrix} = \frac{1}{\sum_{u,v} \mathbf{M}_k^{(i)}(u,v)} \begin{bmatrix} \sum_{u,v} u \mathbf{M}_k^{(i)}(u,v) \\ \sum_{u,v} v \mathbf{M}_k^{(i)}(u,v) \end{bmatrix}, \quad (2)$$

E. Anatomical Landmark Residual Computation Given probe motion, ALRs are defined as the difference between the probe-predicted landmark position and the observed centroid in the subsequent frame, providing a local measure of motion unexplained by probe displacement. Residuals are computed only for valid landmark pairs. Formally, for landmark k with observed centroid $\mathbf{c}_k^{(i)}$ we compute the residuals as described on Algorithm 2

Algorithm 2 Anatomical Landmark Residual (ALR) - Geometric Computation

Input: Observed centroid $\mathbf{c}_k^{(i)}$, probe poses $\mathbf{T}_{\text{probe}}^{(i)}$, $\mathbf{T}_{\text{probe}}^{(i+1)}$, pixel spacing s_x, s_y

Output: Residual $\mathbf{r}_k^{(i)}$ (fetal motion signal)

$\tilde{\mathbf{p}}_k^{(i)} \leftarrow [s_x u_k^{(i)}, s_y v_k^{(i)}, 0, 1]^\top$ ▷ Lift to homogeneous 3-D

$\mathbf{T}_{\text{transport}} \leftarrow (\mathbf{T}_{\text{probe}}^{(i+1)})^{-1} \mathbf{T}_{\text{probe}}^{(i)}$ ▷ Probe motion operator

$\tilde{\mathbf{p}}_{k,\text{pred}}^{(i+1)} \leftarrow \mathbf{T}_{\text{transport}} \tilde{\mathbf{p}}_k^{(i)}$ ▷ Predict under probe motion only

$\mathbf{c}_{k,\text{pred}}^{(i+1)} \leftarrow \text{project_to_pixels}(\tilde{\mathbf{p}}_{k,\text{pred}}^{(i+1)})$ ▷ Back to pixel space

$\mathbf{r}_k^{(i)} \leftarrow \mathbf{c}_{k,\text{obs}}^{(i+1)} - \mathbf{c}_{k,\text{pred}}^{(i+1)}$ ▷ Residual = deviation from probe motion

Small residuals ($\mathbf{r}_k^{(i)} \approx \mathbf{0}$) indicate motion consistent with probe displacement, whereas larger residuals suggest higher fetal motion.

3 Results and Discussion

Preliminary Results FLAME-US comprises 10,000 trajectories generated from 25 fetal head US volumes, totaling approximately 773,000 frames annotated with paired fetal and probe SE(3) motion labels and seven Anatomical Landmark Residuals (ALRs) per frame. Segmentation inference and landmark extraction require 64 ms per frame, while residual computation and pose labeling require 51 ms per frame. With a mean trajectory length of 77 frames, these results demonstrate the scalability of the proposed pipeline for generating large-scale fetal-motion supervision.

Fetal Motion Poses and ALRs Although SE(3) fetal pose labels serve as training supervision, downstream models for motion decomposition and 3-D reconstruction benefit from a more spatially and anatomy-aware signal that does not require explicit fetal pose requirement at inference. We therefore derive ALRs as detailed in Algorithm 2: the pixel-space deviation between the geometrically predicted centroid position at frame $t+1$ predicted under probe motion only, and the observed centroid, such that any residual deviation is attributable solely to fetal movement. Figure 2 confirms this behavior by observing that residuals are negligible under probe-only motion and grow systematically with fetal activity, thereby validating ALRs as a reliable and anatomy-aware fetal motion indicator that correlates with fetal rotation magnitude and supports learnable downstream motion-quantification models

Clinical Reliability of ALRs: Correlation Between Fetal Motion and Residual Motion Figure 3 demonstrates that ALRs strongly indicate fetal head motion. Mean ALRs correlates strongly with fetal rotation magnitude ($r = 0.74$, $p < 0.001$), while probe translation shows only weak independent correlation ($r = 0.29$), suggesting minimal influence on the ALRs signal. ALRs increase consistently with fetal rotation. The segmentation module predicts all 24 anatomical landmarks; however, only reliable structures were retained for

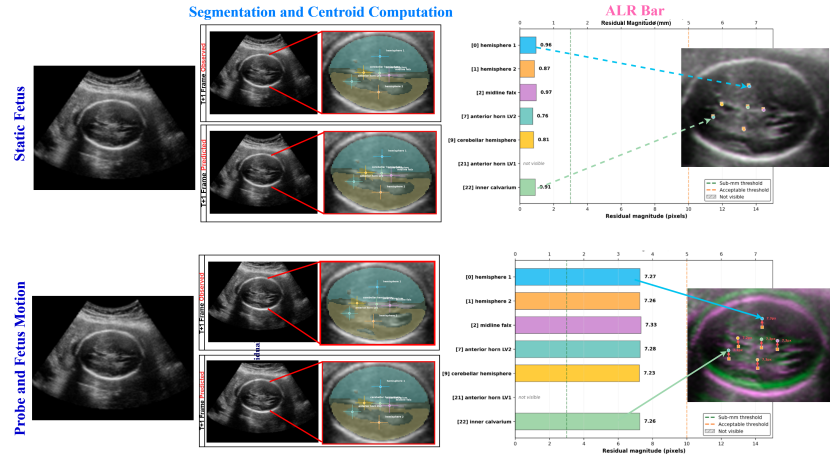


Fig. 2. Qualitative motion analysis across two regimes: static fetus with probe-only motion, and joint fetus-probe motion activity.

motion estimation. To identify robust contributors, each landmark was evaluated across all 773,000 frames based on motion sensitivity and visibility. Motion sensitivity was quantified through correlation between ALRs and fetal rotation magnitude, and only landmarks with strong correlation ($r > 0.74$) were considered. Visibility was assessed using a composite criterion incorporating: (1) high segmentation confidence, (2) strong local anatomical contrast between the predicted mask and its dilated neighbourhood, and (3) low image blur. Based on these measures, landmarks were categorised as *Good* ($\geq 50\%$ visibility), *Partial* (30–50%), or *Poor* ($< 30\%$). Although motion sensitivity remained consistently high across landmarks, several structures exhibited limited visibility. Therefore, only the seven structures (S1–S7) achieving visibility of at least 50% were retained, providing stable and reliable residual estimation.

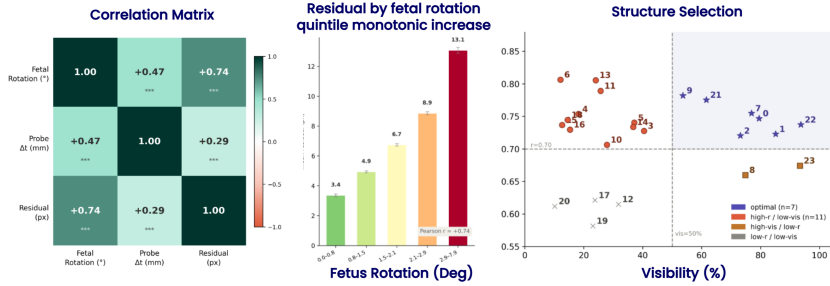


Fig. 3. ALRs as an indicator of fetal head motion on the FLAME-US dataset. *Left:* Pearson correlation matrix showing strong association of ALRs with respect to Probe motion and Fetal Rotation. *Centre:* Mean residual increases consistently across fetal rotation quintiles. *Right:* Selected high-visibility, motion-sensitive landmarks: hemisphere 1–2, midline falx, anterior midline falx, anterior horns of the lateral ventricles 1–2, and inner/outer calvarium (landmarks 0, 1, 2, 7, 9, 21, 22, and 23).

Segmentation Module Validation Accurate ALR estimation depends critically on reliable segmentation quality. To validate the segmentation module, the proposed dual-head U-Net was compared against a vanilla U-Net [16] on STATIC frames from simulated trajectories, in which fetal motion was absent and only probe movement was present. Evaluation was performed on approximately 20 static frames from each of 250 trajectories sampled from the full 10,000-trajectory FLAME-US dataset, where fetal motion was absent and only probe movement was present. In this setting, the ideal landmark residual should be zero or near-zero, as no fetal reorientation occurs.

Table 1. Mean $\pm$ std ALR (px) per anatomical structure. S1=Left Anterior Horn, S2=Right Anterior Horn, S3=Falx Cerebri, S4=Left Cerebral Hemisphere, S5=Right Cerebral Hemisphere, S6=Cavum Septi Pellucidi, S7=Inner Calvarium. Lower is better.

Model	S1	S2	S3	S4	S5	S6	S7	Mean
Vanilla	2.37 $\pm$ 2.92	2.67 $\pm$ 3.15	2.84 $\pm$ 4.04	3.47 $\pm$ 4.55	2.79 $\pm$ 3.70	1.98 $\pm$ 2.15	3.45 $\pm$ 4.30	2.80
Dual-Head	0.89 $\pm$ 0.42	0.88 $\pm$ 0.39	0.79 $\pm$ 0.36	0.75 $\pm$ 0.33	0.79 $\pm$ 0.37	0.32 $\pm$ 0.14	0.92 $\pm$ 0.44	0.76

Table 1 demonstrates that the dual-head model consistently suppresses ALR across all seven anatomical structures compared with the vanilla U-Net (mean: 0.76 px vs. 2.80 px). The elevated ALRs observed with the vanilla U-Net indicate segmentation inconsistencies that perturb landmark centroid localization, producing motion signals despite the absence of true fetal movement. In contrast, the dual-head architecture yields more stable centroid estimates across varying probe orientations, resulting in ALRs that more faithfully reflect fetal motion.

Dataset Utility Benchmarks To validate that FLAME-US-generated SE(3) labels constitute a learnable and discriminative supervision signal for downstream tasks such as motion quantification, we benchmark a fetal-relative pose regression task: given a frame pair (I_i, I_{i+1}) , predict the SE(3) transform of fetal motion alone, isolated from probe motion. We train four baseline architectures on a 250-trajectory subset and evaluate rotational error on this fetal-relative target, with a breakdown by fetal-motion-magnitude quintile. As shown in Table 2, ResNet-18+FiLM achieves the best fetal rotation error (0.438 rad), 24.6% lower than ResNet-18 (0.581 rad), with error decreasing sharply from Q1 to Q4 (1.060 $\rightarrow$ 0.145 rad) and a marginal uptick at Q5 (0.152 rad), confirming that the generated labels encode a learnable, motion-magnitude-aware signal.

Table 2. Baseline comparison on FLAME-US test set, including probe/fetal motion errors and fetal rotation across motion-displacement quintiles(Q1–Q5).

Model	Params	probe_r	fetal_r	fetal_t	Q1	Q2	Q3	Q4	Q5
ResNet-18	23.0M	0.641	0.581	1.516	1.080	0.975	0.415	0.231	0.209
ResNet-18+FiLM	23.9M	0.582	0.438	0.960	1.060	0.633	0.205	0.145	0.152
EffNet-B0	9.5M	0.657	0.980	1.955	1.241	1.276	0.860	0.675	0.833
EffNet-B0+FiLM	10.7M	0.563	0.721	1.183	1.102	0.741	0.591	0.724	0.621

4 Conclusion and Future Work

FLAME-US introduces a scalable framework for generating disentangled fetal and probe motion annotations from patient-derived 3D US Volumes. The proposed ALRs provide a compact representation of fetal-induced anatomical displacement and exhibit a strong association with fetal motion magnitude ($r = 0.74$, $p < 0.001$), while baseline experiments demonstrate that the generated labels contain learnable motion information. Together, these findings highlight the potential of simulation-based supervision for advancing motion-aware ultrasound analysis. Future work will extend the framework to additional fetal anatomies and investigate translation toward clinical video-based fetal–probe motion disentanglement and downstream motion-aware ultrasound applications.

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